Ex. 3

Practical 8

AIM: To study the Packet tracer tool Installation and User Interface Overview

To understand environment of CISCO PACKET TRACER to design simple network

INTRODUCTION:

A Simulator, as the name suggests, simulate network devices and its environment. Packet Tracer is an exciting network design, simulation and modelling tool.

1. It allows you to model complex systems without the need for dedicated equipment.
2. It helps you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device
3. It is available for both the Linux and windows desktop environment.
4. Protocols in packet Tracer are coded to work and behave in the same way as they would on real hardware

INSTALLING PACKET TRACER

To download Packet Tracer, go to https://www.netacad.com and log in with your Cisco networking academy credentials; then click on the Packet Tracer graphic and download the package appropriate four your OS

Windows

Installation in Windows is pretty simple and straightforward, the setup comes in a single file named PacketTracer_Setup6.0.1.exe. Open this file to begin the setup wizard, accept the license agreement, choose a location and start the installation

Linux

Linux users with an Ubuntu/Debian distribution should download the file for Ubuntu and those using Fedora/Redhat/CentOS must download the file for Fedora. Grant executable permission to this file by using chmod and execute it to begin the installation

chmod +x PacketTracer601-i386-installer-rpm.bin

./PacketTracer601-i386-installer-rpm.bin

USER INTERFACE OVERVIEW

The layout of Packet Tracer is divided into several components. The components Packet Tracer are as follows: follow the numbering with explanation

1. Menu-bar: This is a common menu found in all software applications; it is used to open, save, print change preferences and so on

2. Main Toolbar: This bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo and redo and on the right hand side is an icons for entering network information for the current net

3. Logical / Physical workspace Tab :

These tabs allow you to toggle btwn the logical and Physical work areas.

4. workspace : This is the area where topologies are created and simulation are displayed

5 common Tools Bar : This toolbar provides controls for manipulating topologies such as select, move, layout, place note delete, inspect, resize shape & complex PDU

6. User created packet box = Users can create highly customised packets to test their topologies from this area, & the results are displayed as a list

7. Real-time Simulation tabs :

These tabs are used to toggle btwn the real and simulation modes. Buttons are also provided to control the time and to capture the packets

8. Network component box :

This component contains all of the network and end devices available with Packet Tracer, and is further divided into two areas :

Area 7a : Device - type selection box - This area contains device categories Area 7b : Device - Specific selection box - when a device category is selected, this selection box displays the different device models within the category

d) Analyse the behaviour of network devices using Packet Tracer Simulator :-

1. From the network component box, click and drag & drop the below components
   a) 4 Generic PCs & One hub
   b) 4 Generic PCs & One switch

2. Click on connections
   a) click on copper straight - Through cable,
   b) Select one of the PC and connect it to hub using the cable. The link LED should glow in green, indicating that the link is up. Similarly connect remaining 3 PCs to the HUB.
   c) Similarly connect 4 PCs to the Switch using copper - straight through cable

3. Click on the PCs connected to HUB, go to the desktop tab, click on IP configuration, and enter an IP address and subnet mask. HOSTS, the default gateway and DNS server information is not needed as there are only two end devices in the network

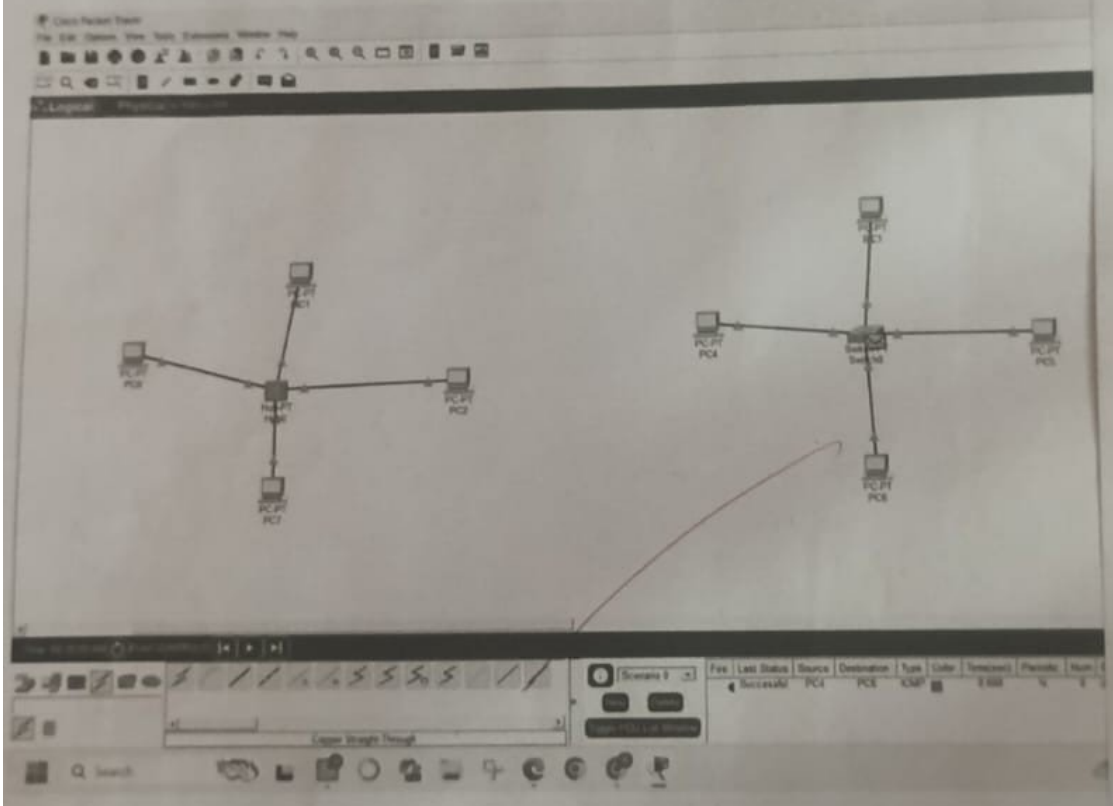
4. Click the PDU (message icon) from the common tool bar
   a) Drag and drop it on one of the PC (source machine) and then drop it on another PC (destination machine) connected to the hub

5. Observe the flow of PDU from source PC to destination PC by the selecting Realtime mode of simulation

5. Observe the flow of PDU

6. Repeat step #3 to step #5 for the PCs connected to the Switch

7. Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviour of Switch and Hub



Student Observation:

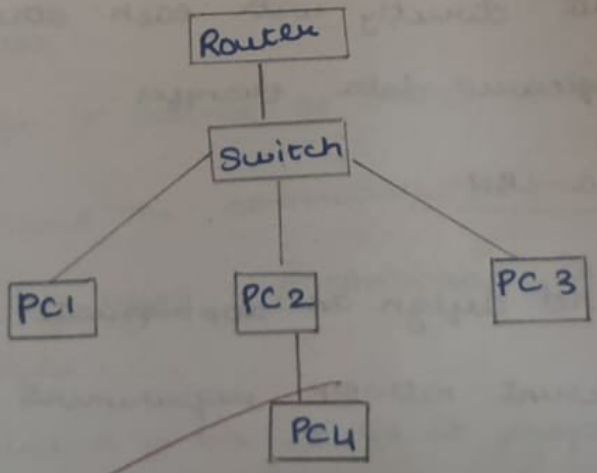
a) From your observation write down the behaviour of switch and HUB in terms of forwarding packeted recieved by them

Switch: Recieves a packet and forwards it only to the intended destination port using the MAC address.

HUB: Recieves a packet and broadcasts to all connected ports

b) Find out the network topology implimented in your college and draw and label that topology in your observation book.

Our college network mainly uses Star Topology, where all computers are connected to a central switch.



The network topology observed is Star Topo